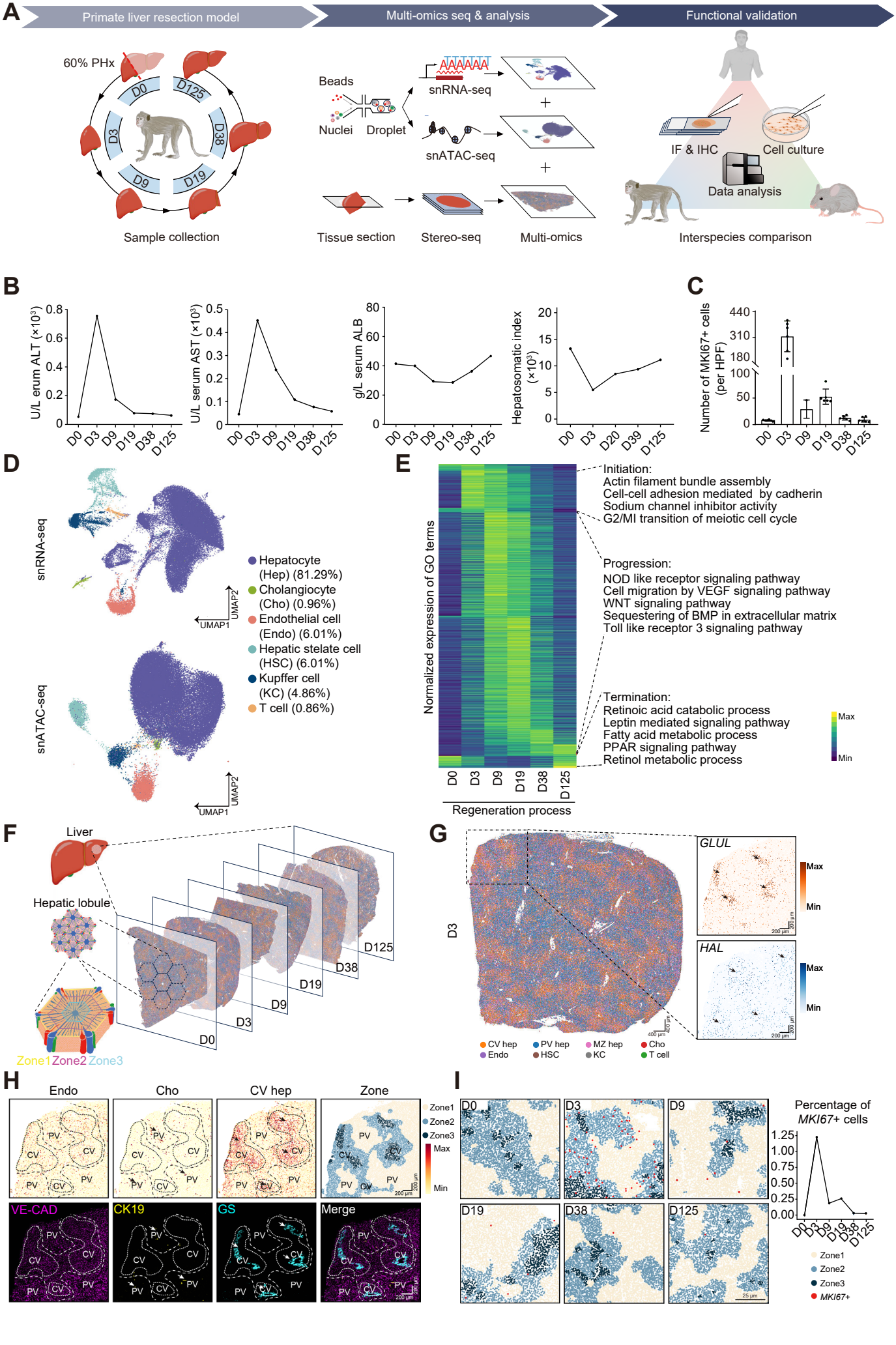

A
Primate liver resection model
Multi-omics seq & analysis
Functional validation
60% PHx
D0
D3
D9
D19
D38
D125
Sample collection
Beads
Nuclei
Droplet
snRNA-seq
snATAC-seq
Tissue section
Stereo-seq
Multi-omics
IF & IHC
Cell culture
Data analysis
Interspecies comparison
B
U/L erum ALT (×10³)
U/L serum AST (×10³)
g/L serum ALB
Hepatosomatic index (×10³)
C
Number of MKI67+ cells (per HPF)
D
snRNA-seq
snATAC-seq
UMAP1
UMAP2
Hepatocyte (Hep) (81.29%)
Cholangiocyte (Cho) (0.96%)
Endothelial cell (Endo) (6.01%)
Hepatic stelate cell (HSC) (6.01%)
Kupffer cell (KC) (4.86%)
T cell (0.86%)
E
Normalized expression of GO terms
Initiation:
Actin filament bundle assembly
Cell-cell adhesion mediated by cadherin
Sodium channel inhibitor activity
G2/MI transition of meiotic cell cycle
Progression:
NOD like receptor signaling pathway
Cell migration by VEGF signaling pathway
WNT signaling pathway
Sequestering of BMP in extracellular matrix
Toll like receptor 3 signaling pathway
Termination:
Retinoic acid catabolic process
Leptin mediated signaling pathway
Fatty acid metabolic process
PPAR signaling pathway
Retinol metabolic process
Max
Min
Regeneration process
F
Liver
Hepatic lobule
Zone1 Zone2 Zone3
G
D3
GLUL
HAL
CV hep
PV hep
MZ hep
Cho
Endo
HSC
KC
T cell
H
Endo
Cho
CV hep
Zone
VE-CAD
CK19
GS
Merge
I
Percentage of MKI67+ cells
MKI67+